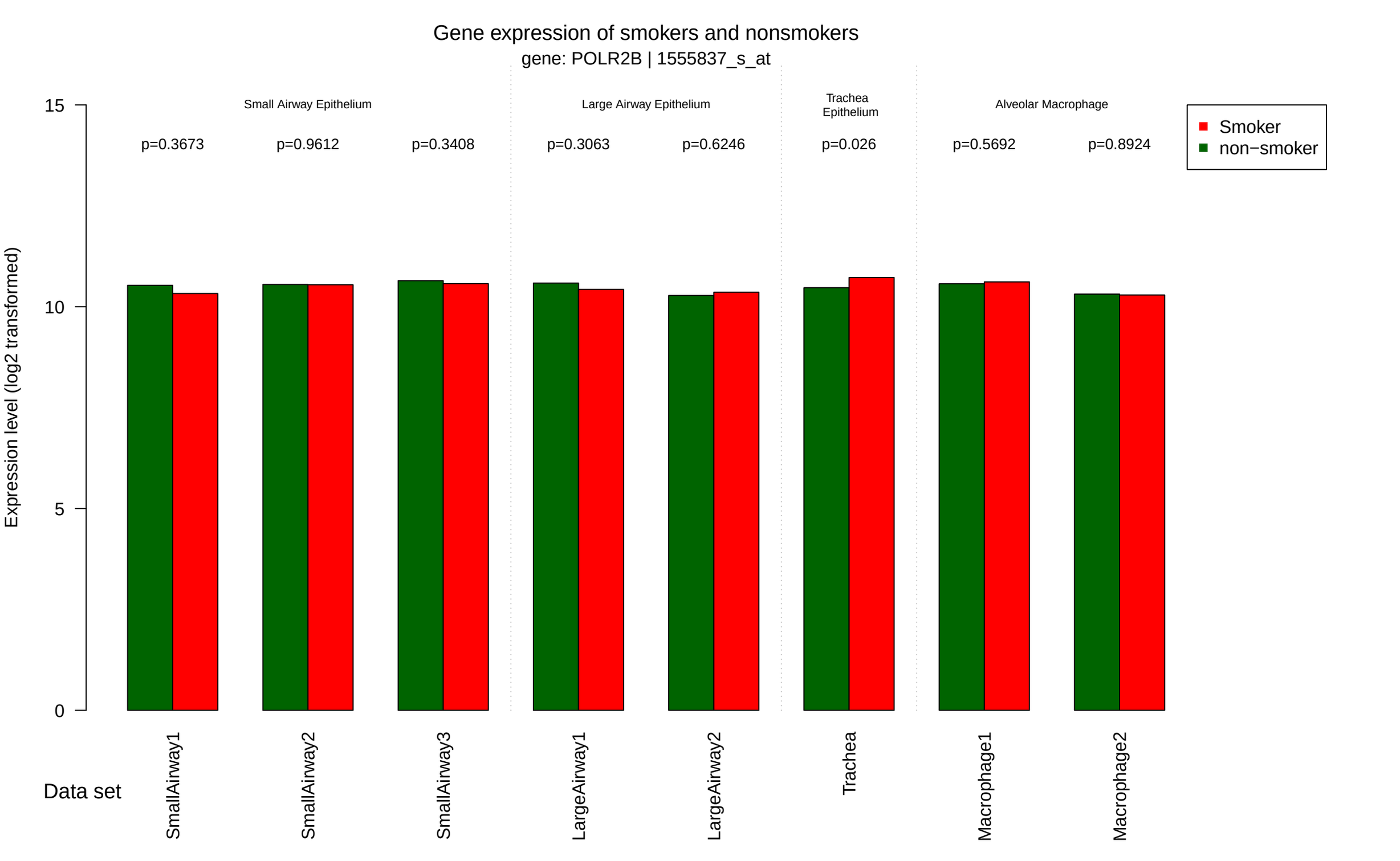
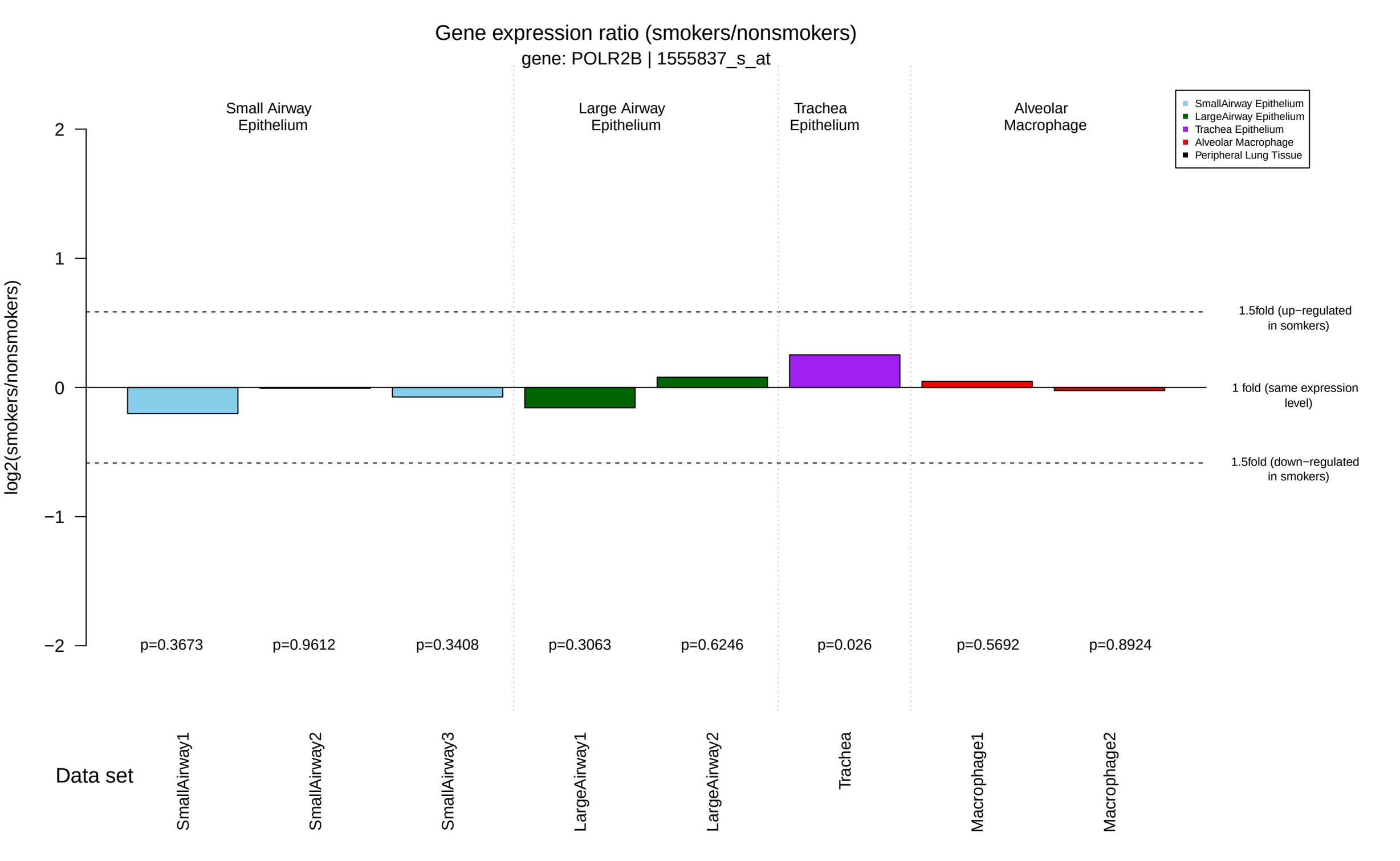
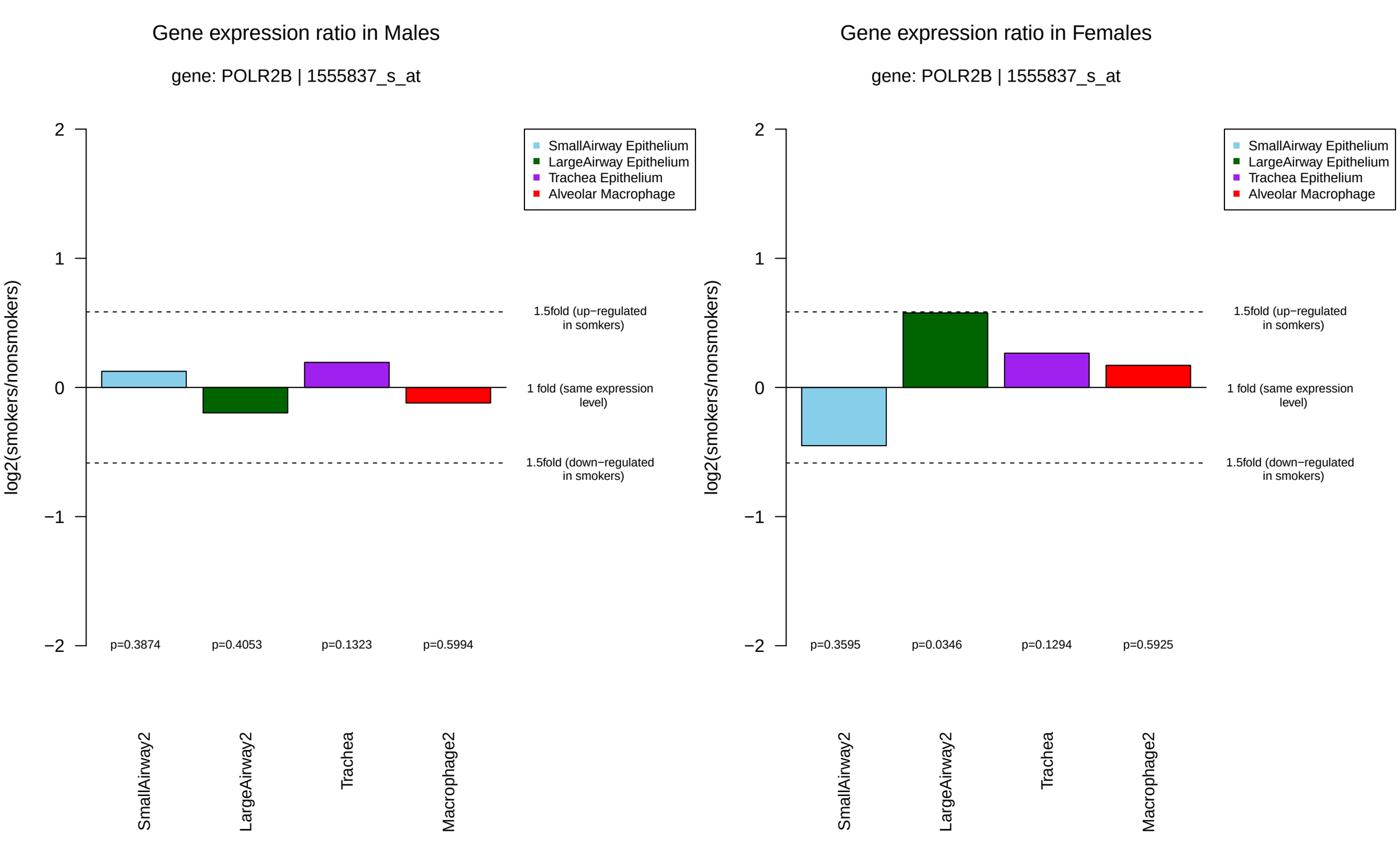
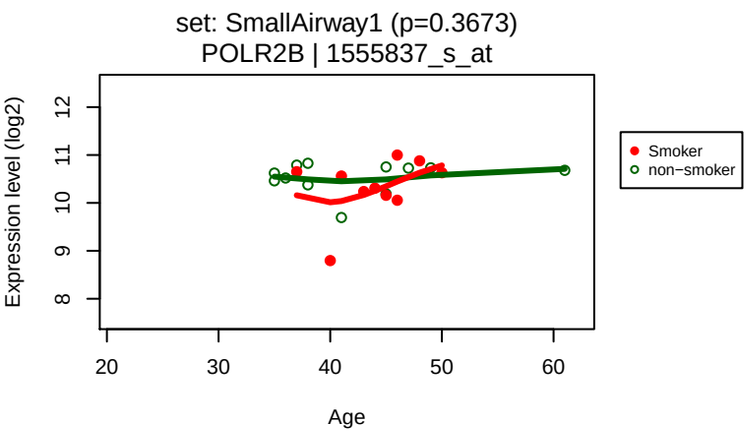
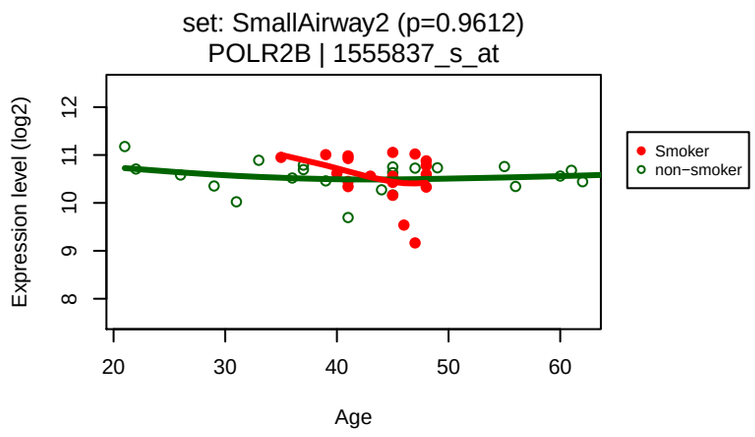
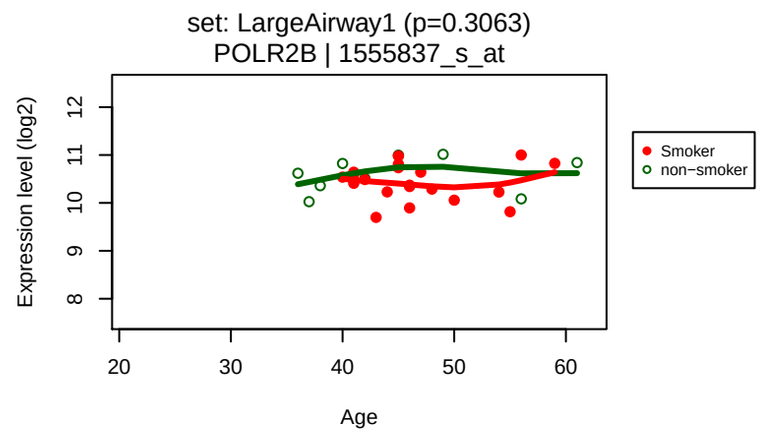

Gene expression of smokers and nonsmokers
gene: POLR2B | 1555837_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3673
p=0.9612
p=0.3408
p=0.3063
p=0.6246
p=0.026
p=0.5692
p=0.8924
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

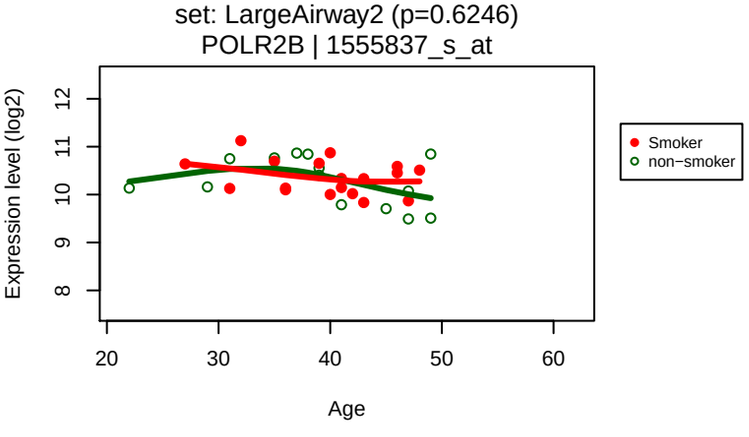

Gene expression ratio (smokers/nonsmokers)
gene: POLR2B | 1555837_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3673
p=0.9612
p=0.3408
p=0.3063
p=0.6246
p=0.026
p=0.5692
p=0.8924
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

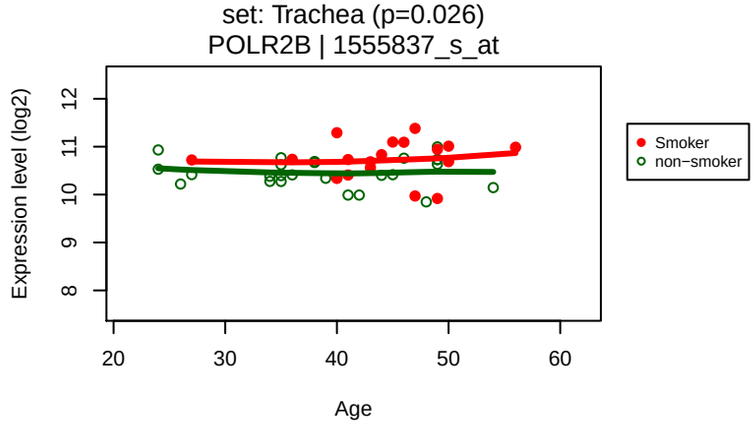

Gene expression ratio in Males
gene: POLR2B | 1555837_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3874
p=0.4053
p=0.1323
p=0.5994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2B | 1555837_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3595
p=0.0346
p=0.1294
p=0.5925
SmallAirway2
LargeAirway2
Trachea
Macrophage2

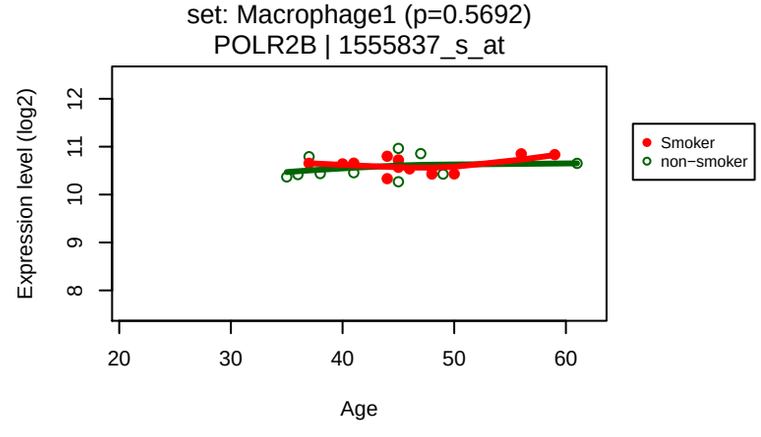

set: SmallAirway1 (p=0.3673)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker

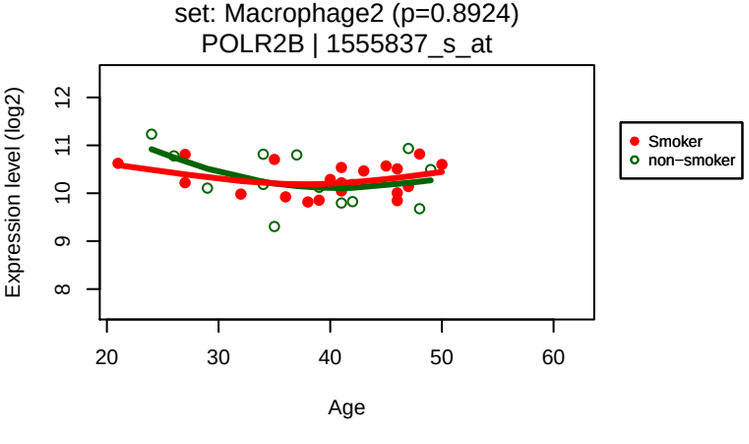

set: SmallAirway2 (p=0.9612)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker

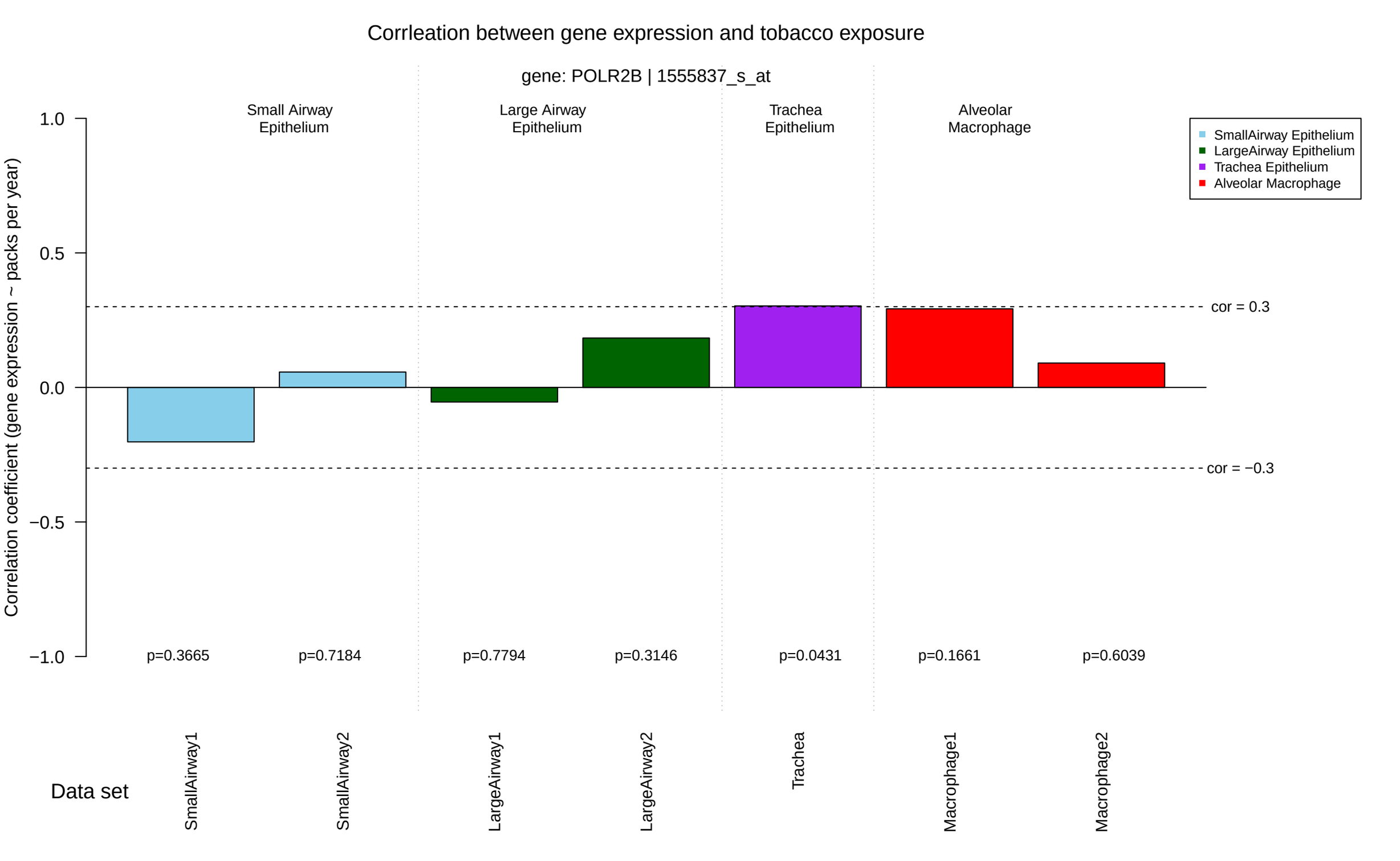

set: LargeAirway1 (p=0.3063)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6246)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.026)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5692)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8924)
POLR2B | 1555837_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2B | 1555837_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3665
p=0.7184
p=0.7794
p=0.3146
p=0.0431
p=0.1661
p=0.6039
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2